



Switching

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Why Switching is required?

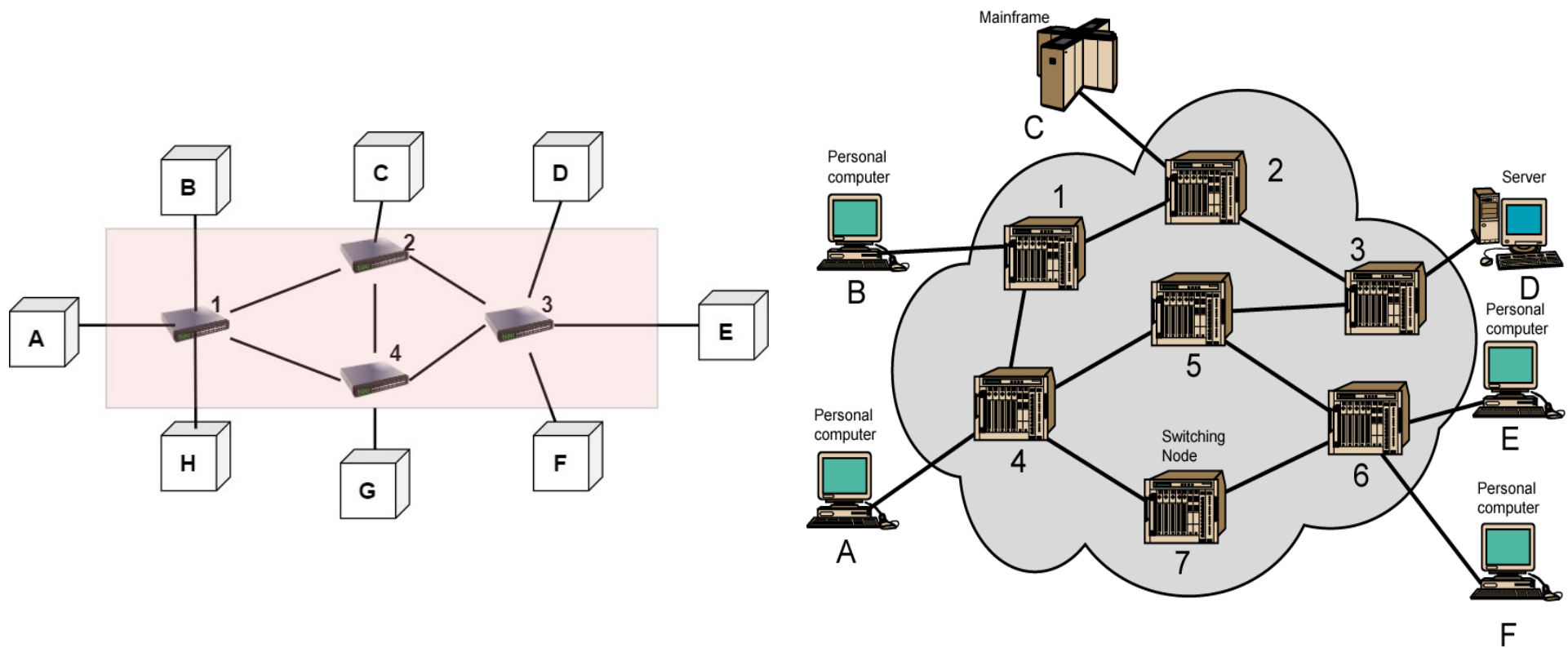
- We can't afford to have **one-to-one connection** between each pair of device, or between server and all other devices in a network
- Moreover, one-to-one connection is an **wasteful** approach w.r.t resources
- A cost-effective solution is required



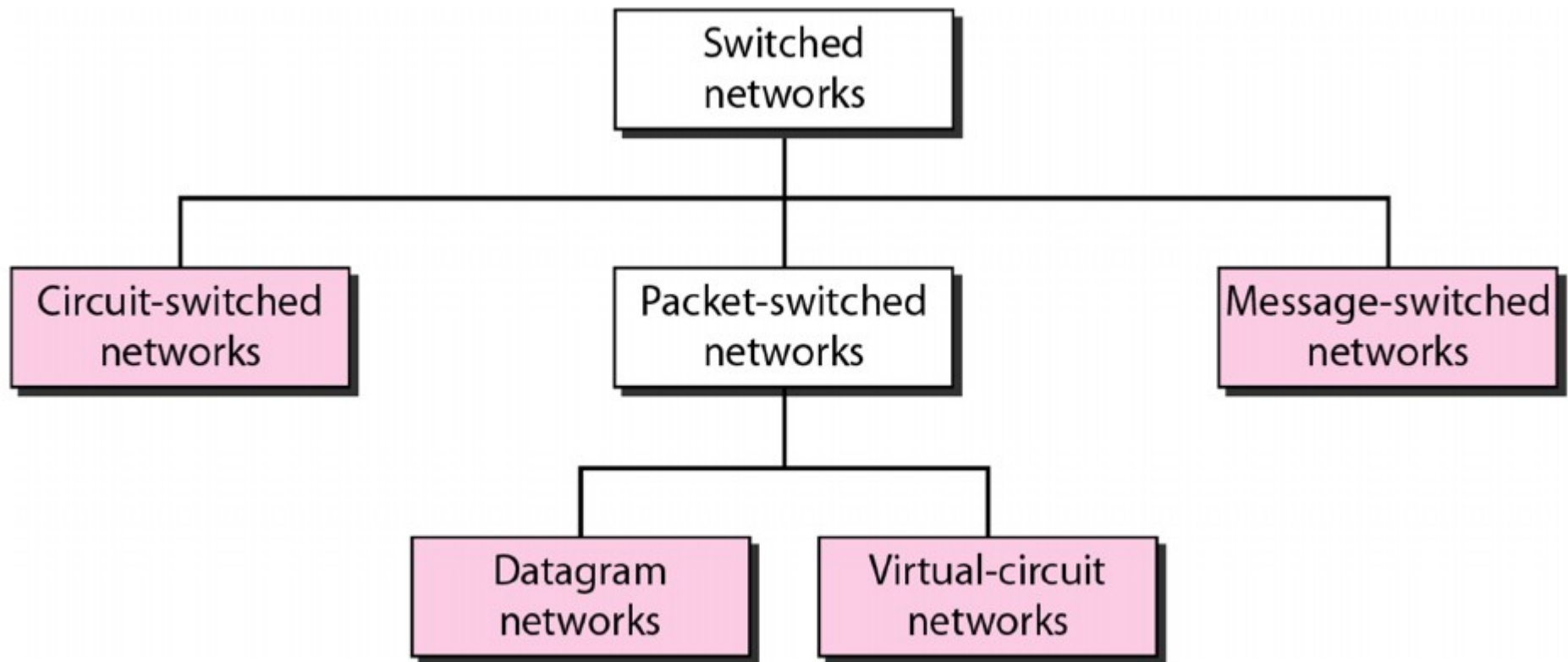
Switching

- A switched network consists of a series of interlinked nodes, called **switches**.
- Switches are devices capable of creating **temporary** connections between two or more devices linked to the switch.
- Some of these nodes are connected to the end systems and rest are used only for routing

Switched network



Switching Methods





Switching and TCP/IP

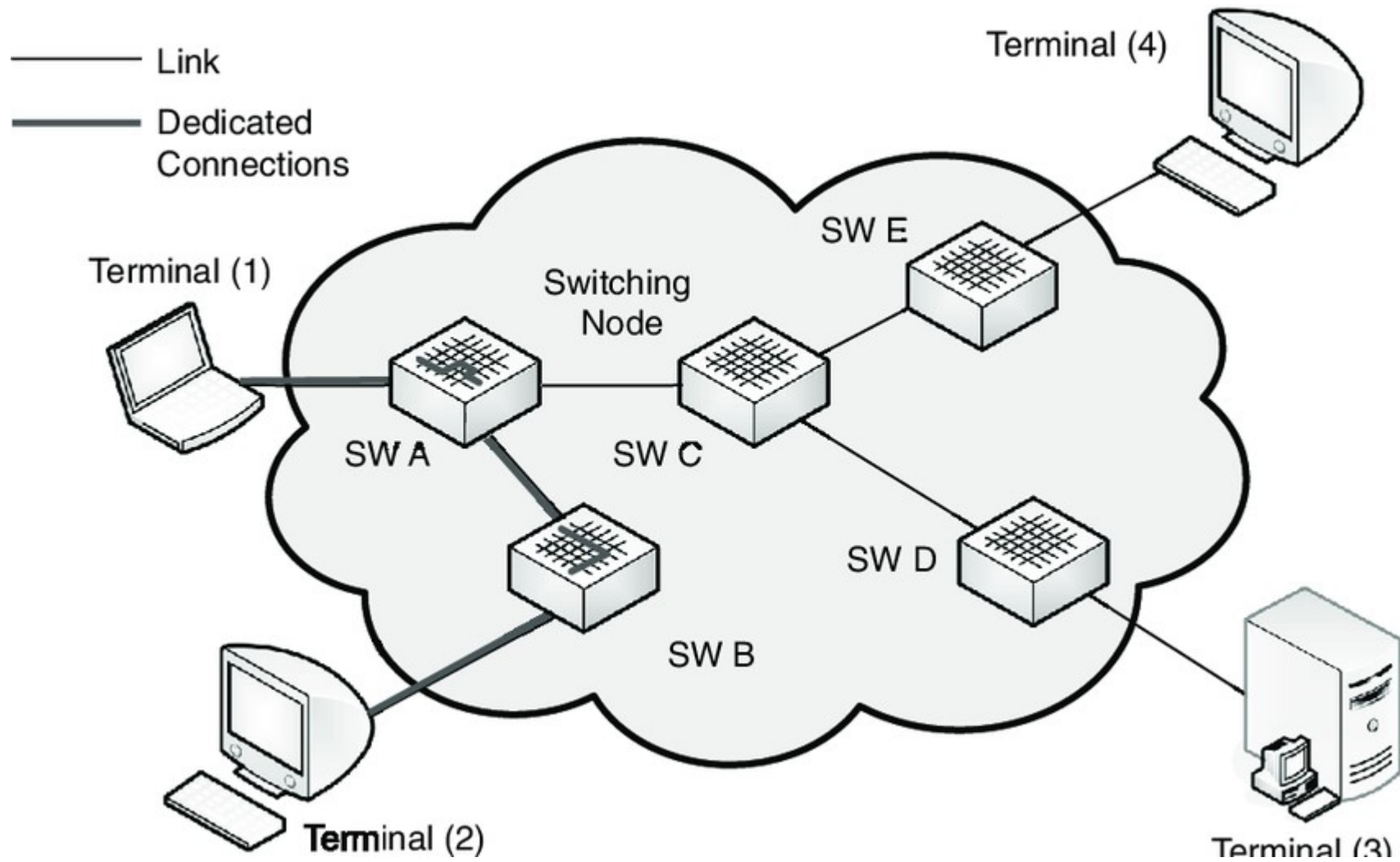
- Physical Layer – Circuit Switching
- Data Link and Network Layer – Packet Switching



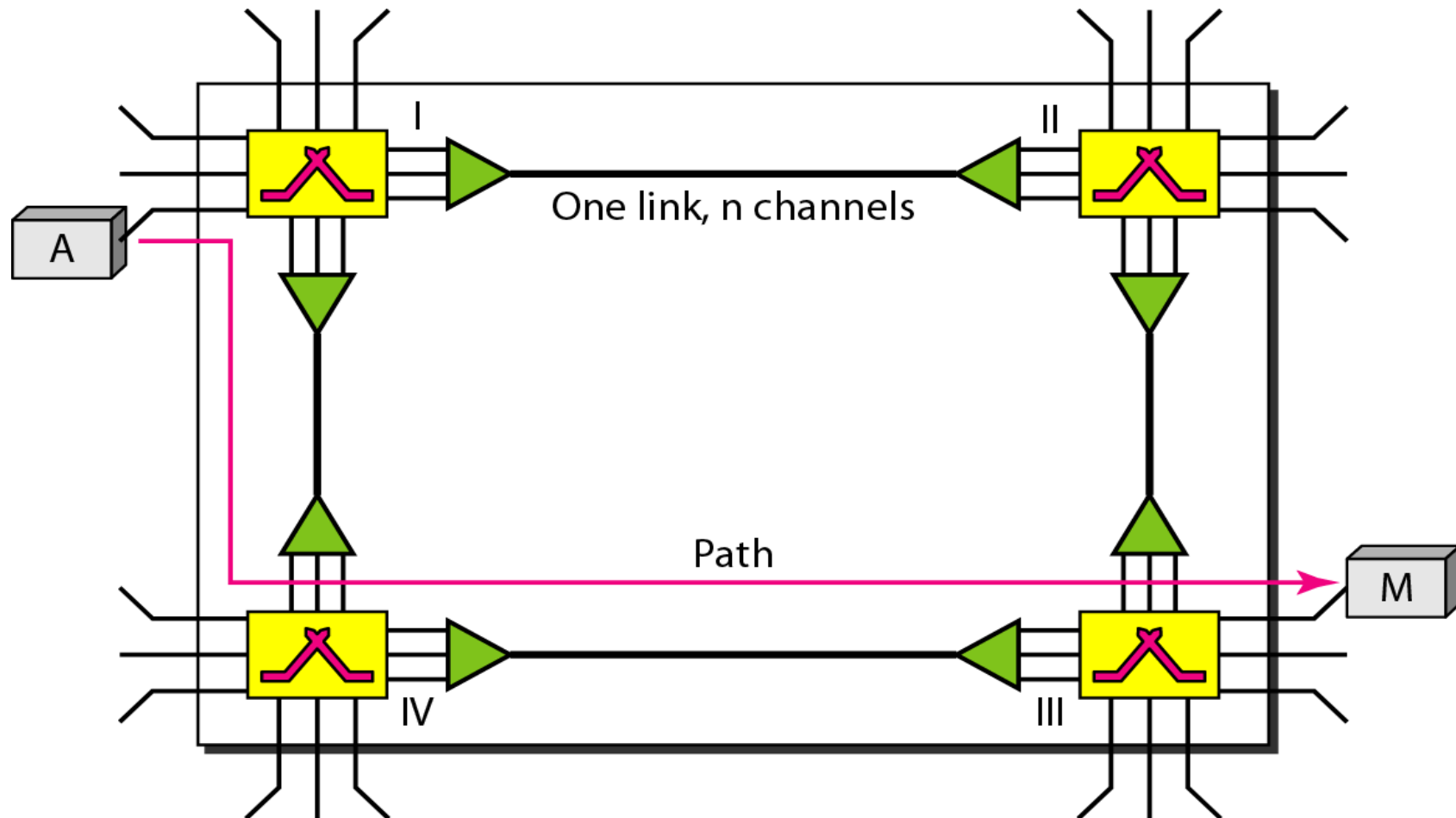
Circuit-Switched Networks

- A circuit-switched network consists of a **set of switches** connected by physical links
- A connection between two stations is a **dedicated** path made of one or more links
- However, each connection uses only one dedicated channel on each link
- Each link is normally divided into n channels by using FDM or TDM.

Circuit-Switched Networks



Circuit-Switched Networks





Important point about circuit-switching

- It takes place at the **physical layer**
- Before starting communication, the stations must make a **reservation** for the resources to be used dedicatedly during the communication
 - channels (bandwidth in FDM and time slots in TDM)
 - switch buffers
 - switch processing time
 - switch input/output ports
- No packeting, data flows in a continuous manner



Phases of Circuit-Switching

- Setup phase
 - Creation of **dedicated channels** between the switches
 - Making connection (An Acknowledgement from receiver)
- Data transfer phase
- Teardown phase
 - Signal sent to each switch to release the resources

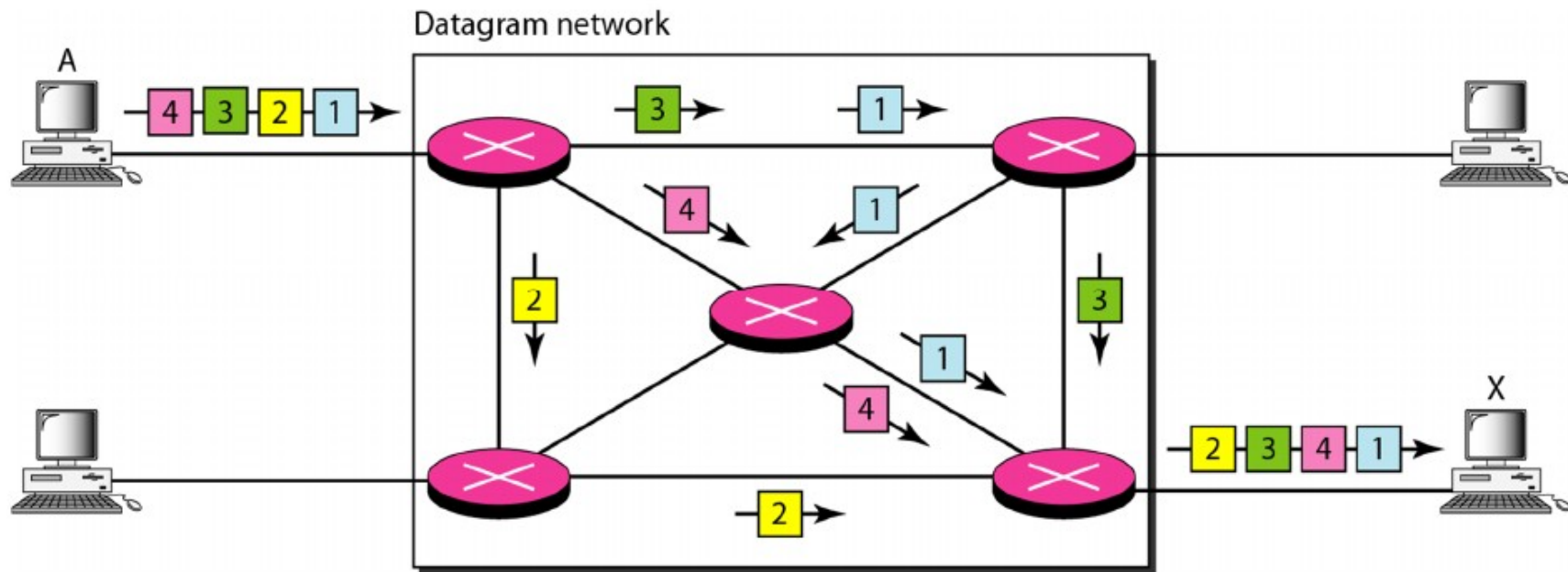


Packet Switching

- Communication in term of **packets**
- **No prior** resource allocation
- Infact, resource allocated on **demand**
- Sometime packet has to **wait** when other packets are being processed
- **Higher delay** compared to circuit switched n/w
- Two types
 - Datagram network
 - Virtual circuit network

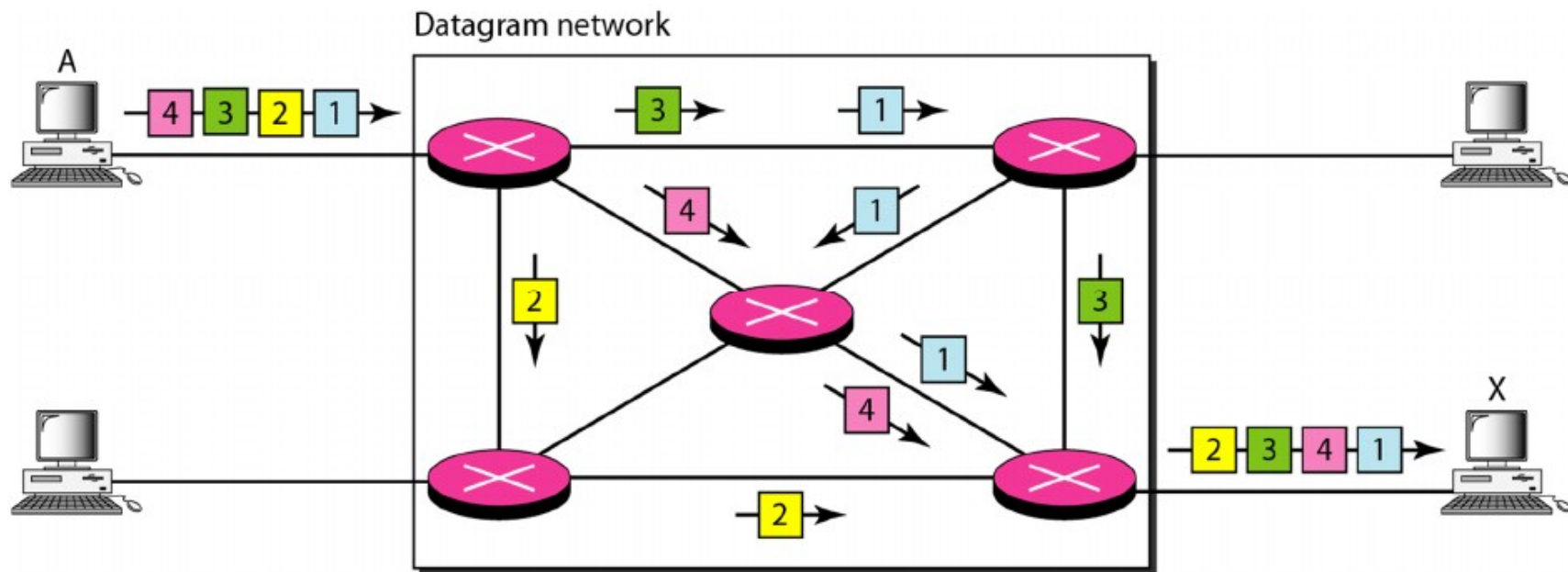
Datagram Network

- Each packet is treated independently of all others
- Packets are referred to as **datagrams**.
- Datagram switching is normally done at the network layer.



Datagram Network

- Also referred as connection less networks
- No setup or teardown
- A forwarding table may required at each switch
- A crucial field in the header of the datagram is the **destination address** that helps switch to decide the port through with packet should be forwarded

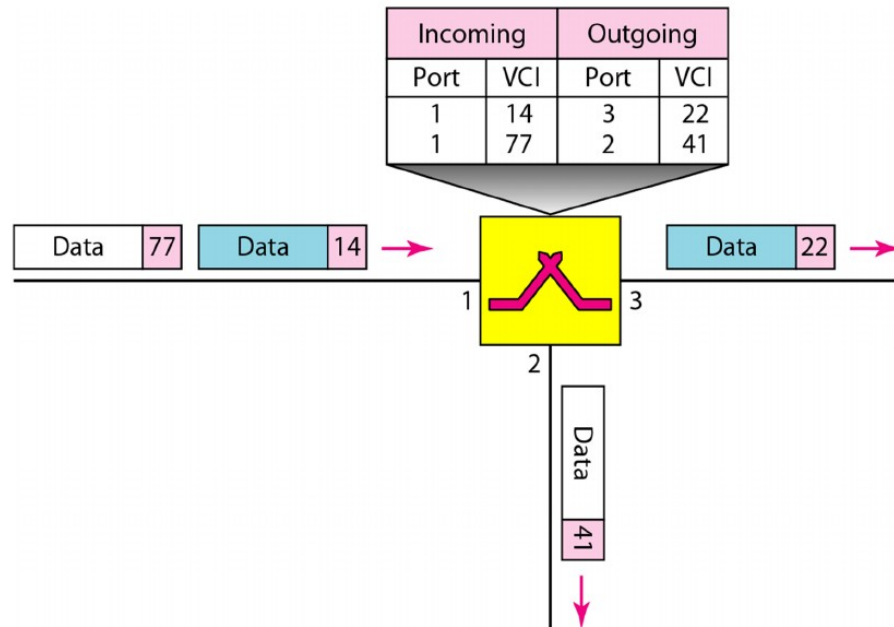




Virtual Circuit Networks (VCN)

- A hybrid of circuit switched and datagram network
- Normally, implemented in datalink layer
- Two types of addressing are used
 - Global Addressing
 - Unique in the network or international network scope
 - **Virtual Circuit Identifier** (local addressing)
 - Scope is limited to the link between switches
 - VCI changes when a frame leave a switch

Components of VC



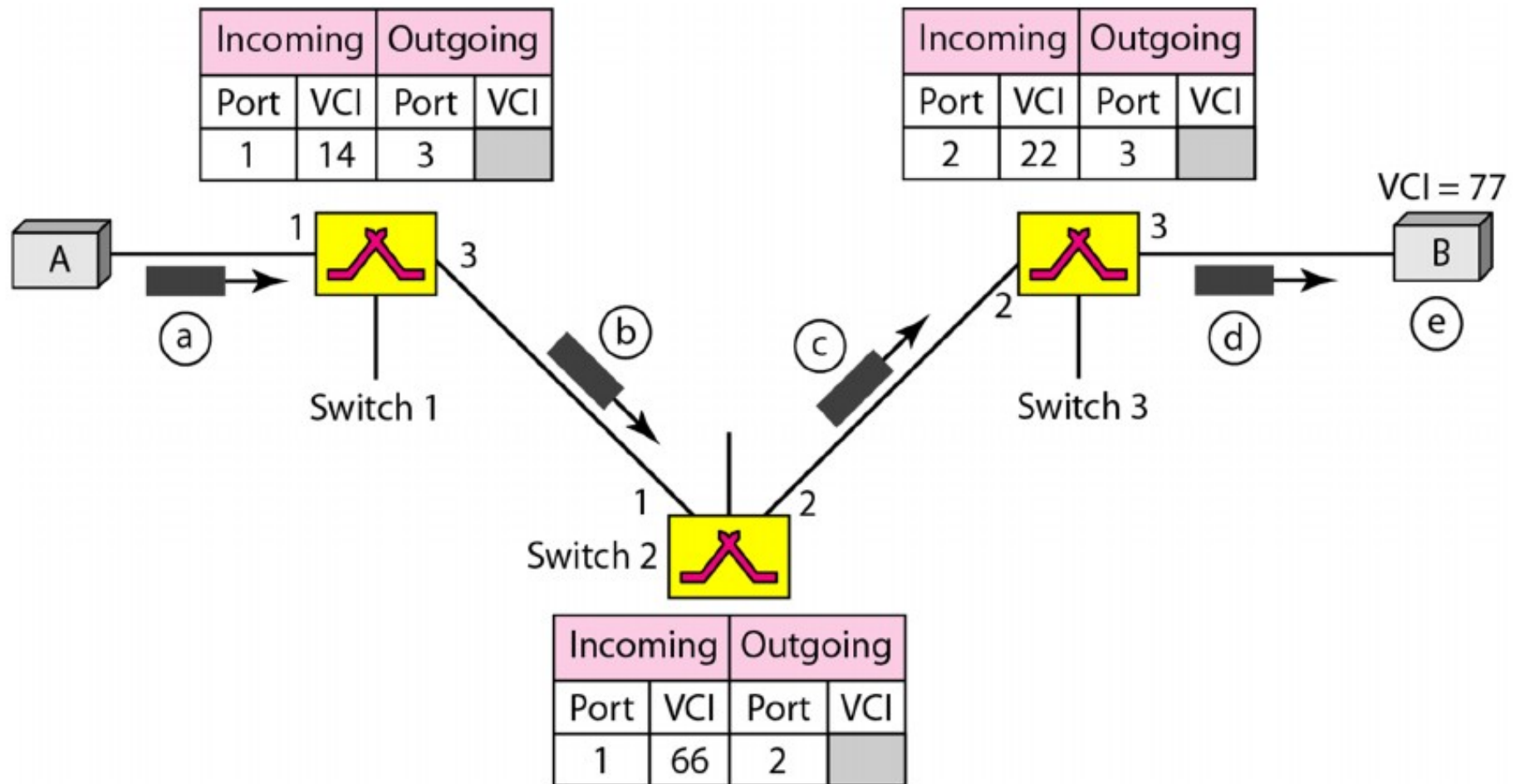
- A path between source and destination
- VCI
- Entries in the forwarding table of switches



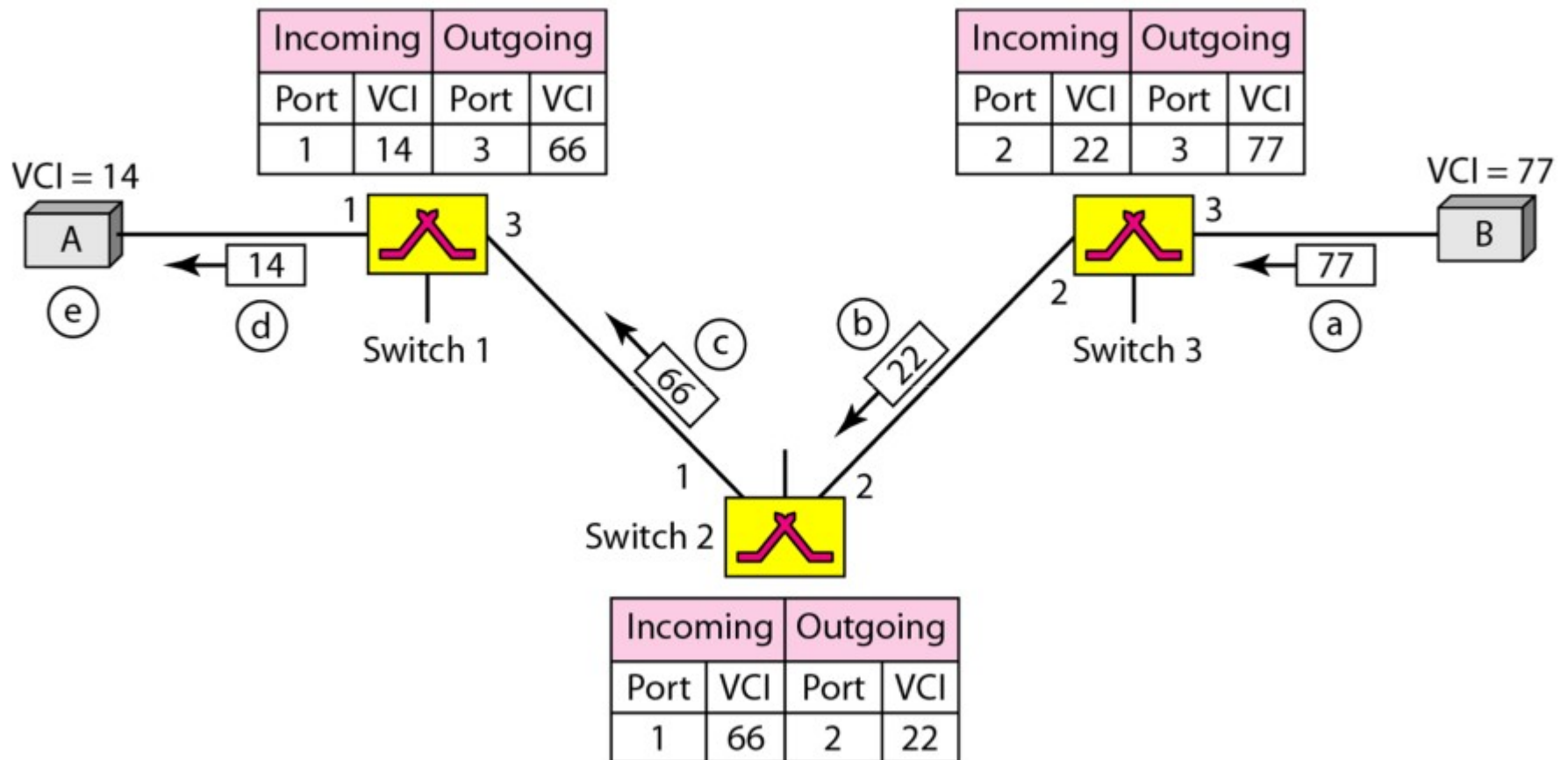
VCN Phases

- Setup phase
 - Sending device contacts its n/w layer for the **global address** of the receiver
 - n/w layer determines the **path to the receiver**
 - Link layer determines the **VCI for each link**
 - Adding of entries in the **forwarding table** of each switches
 - Resource reservation

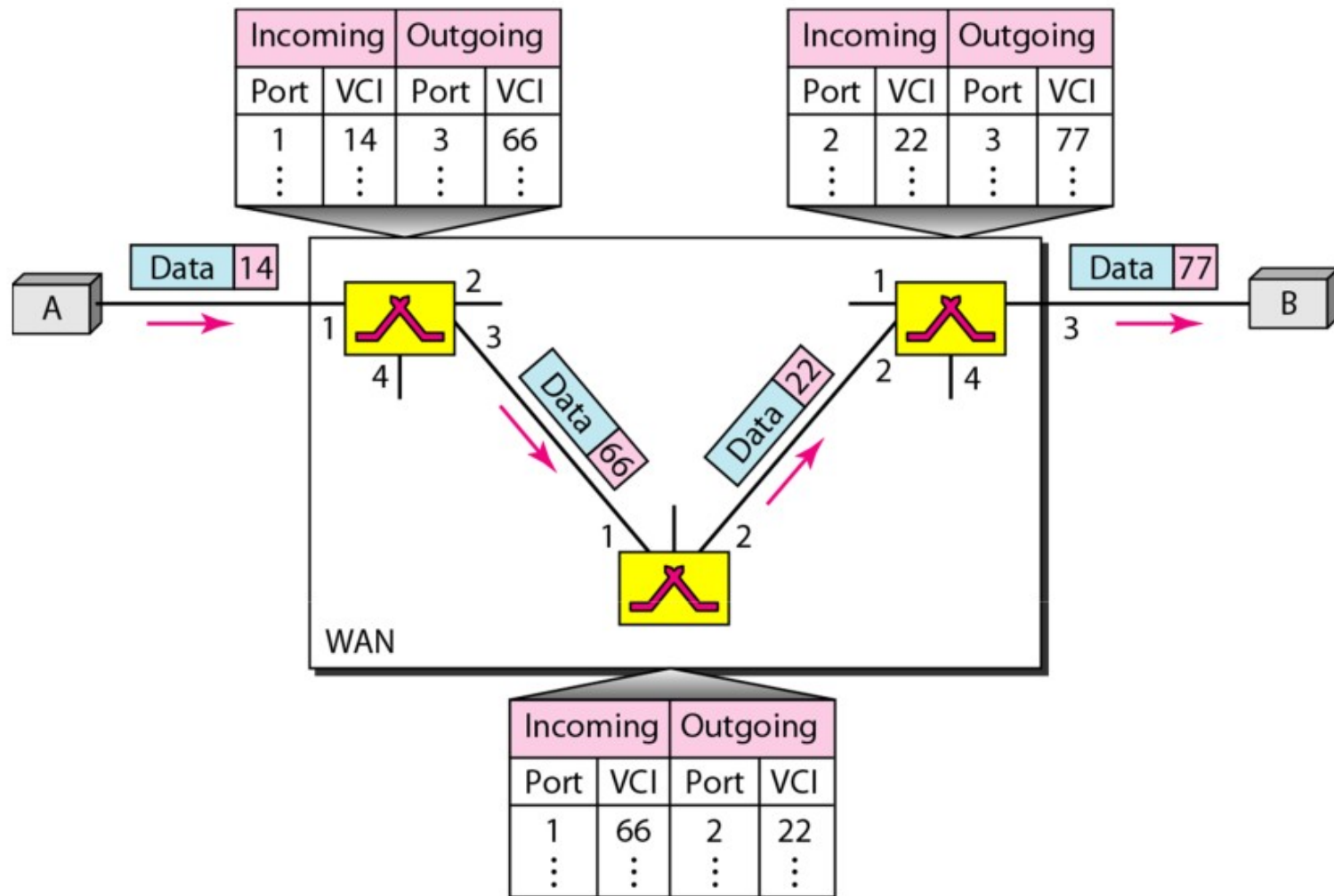
Setup request



Setup Acknowledgement



Data transfer phase





VC teardown phase

- Source sends a special frame called a **teardown request**
- Destination responds with a teardown **confirmation** frame
- All switches **delete** the corresponding entry from their tables.



Comparison

	Efficiency	Delay
• Circuit Switched	Low	Low
• Packet Switched	High	High
• Virtual Circuit	High	One Time Delay



For Self Study – Structure of the Switches

- Crossbar Switch
- Multistage Switch
- Time-division Switch



Next : Data Link Layer